



BAYU LIBRARY

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User Guide — System Flow for Members and Librarians

Bayu Library Management System · ICT502 Database Engineering · Group 2, NBSC2306A

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1. System Overview

What Bayu Library is and who uses it.

Bayu Library Management System is a web-based library management system that allows **librarians (staff)** to manage the book collection, members, and borrowing transactions, and allows **members** to check the status of their own borrowings online.

There are **two user roles** that sign in through separate pages:

Role	Login Page	Credentials	Main Function
Librarian (Staff)	/login	Staff ID + Password	Manage the book catalogue, members, borrowing/returning, and (for certain ranks) other staff accounts
Member	/member/login	Email + Password	Check their own borrowing history, status (borrowed/overdue/returned), and any outstanding fines

Important note: A Member account cannot be self-registered — it is created by a librarian through the **Members** page in the dashboard. Likewise, borrowing/returning a book is handled by a librarian at the counter (Borrowings page), not by the member as a self-service action. This mirrors a physical, counter-based library model rather than a self-checkout system.

2. Usage Flow — Member

What a member can do after signing in.

① **Open /member/login**

A dedicated sign-in page for members (separate from the staff page).



② **Sign in with Email + Password**

These credentials are issued by a librarian when the membership is registered.



③ **"My Borrowings" is displayed**

A list of every book ever/currently borrowed — book title, borrow date, due date, status (**BORROWED** / **OVERDUE** / **RETURNED**), and the fine amount (RM) if overdue.



④ **Logout**

End the session via the Logout button in the header.

Member scope: This portal is *read-only self-service* — a member can only view their own records. There is currently no book reservation, self-checkout, password change, or profile update function.

3. Usage Flow — Librarian

The staff workflow from signing in to handling day-to-day transactions.

① **Open /login and sign in with Staff ID + Password**

Example Staff ID format: STF-2020-001.



② **Dashboard**

A summary of statistics: total Categories, Authors, Members, Librarians, Books, Total Borrowings, and Active Borrowings. Each card links directly to the related page.



③ **Manage Master Data**

Categories (book categories & parent/child relationships), **Authors**, and **Books** (title, ISBN, category, authors, total/available copies) — can be Added and Edited. Delete is **Management Only**.



④ **Manage Members**

Register new members (Member ID & Person ID are auto-generated), update their details, and set the membership type — **STANDARD** (14-day loan period) or **PREMIUM** (30 days). Delete is **Management Only**.



⑤ **Manage Librarians (Staff)**

View the full staff list (visible to every rank). Add / Edit / Delete a staff account is **Management Only** — regular staff only see "View only".



⑥ **Borrowings — Borrow & Return**

New Borrowing: pick the Member, the Book (only those with copies available), and the Librarian processing it; the borrow date & due date are calculated automatically based on the membership type (can still be adjusted manually).

Return: click the "Return" button — the system automatically calculates a fine of **RM1.00/day late** if the return date is past the due date.



⑦ **Change Password & Logout**

Both functions are available to every staff member at the bottom of the sidebar.

4. Access Levels by Position (Permission Tiers)

As of the latest update, a staff member's position now determines their actual access in the system.

● Management — Head / Senior Librarian ● Staff — Librarian / Assistant Librarian

Action	Management (Head / Senior Librarian)	Staff (Librarian / Assistant Librarian)
View all pages (Categories, Books, Members, etc.)	✓ Yes ✓ Yes	
Add / Edit Categories, Authors, Books, Members	✓ Yes ✓ Yes	
Delete Categories, Authors, Books, Members	✓ Yes	X No
Process book borrowing & returns (Borrowings)	✓ Yes ✓ Yes	
Add / Edit / Delete other Librarian accounts	✓ Yes	X No (View only)
Change own password	✓ Yes ✓ Yes	

Why this design? Actions that are hard to reverse (deleting records) or security-sensitive (managing other staff accounts) are restricted to Head/Senior Librarians only — following the *least privilege* principle commonly used in real-world systems. Regular staff can still carry out their daily work (registering members, adding books, processing borrowings) without any restriction.

How it's determined: The system checks the staff's *Position* text field (case-insensitive) — if it contains the word **"Head"** or **"Senior"**, that staff member is treated as Management. Anything else (including unrecognized titles) is automatically treated as Staff (fail-safe).